Topic: Exact equations

Question: Which equation is an exact differential equation?

Answer choices:

A $(3x^2y - 3y^2)\,dx + (x^3 + 9xy^2)\,dy = 0$

B $(3x^2y - 3y^3)\,dx + (x^3 - 9xy^2)\,dy = 0$

C $(3x^2y - 5y^3)\,dy - (x^3 + 9xy^2)\,dx = 0$

D $(3x^2y - 3y^3)\,dy + (2x^2 + 9xy^2)\,dx = 0$

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Solution: B

If we start with answer choice B, we can identify M and N as

$$M = 3x^2y - 3y^3$$

$$N = x^3 - 9xy^2$$

If we find $\partial M/\partial y$ and $\partial N/\partial x$,

$$\frac{\partial M}{\partial y} = 3x^2 - 9y^2$$

$$\frac{\partial N}{\partial x} = 3x^2 - 9y^2$$

Because these partial derivatives are equal,

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

answer choice B is an exact differential equation.

We can use the same process to show that answer choice A is not an exact equation,

$$\frac{\partial}{\partial y}(3x^2y - 3y^2) \neq \frac{\partial}{\partial x}(x^3 + 9xy^2)$$

$$3x^2 - 6y \neq 3x^2 + 9y^2$$

that answer choice C is not an exact equation,

$$\frac{\partial}{\partial y}(-(x^3 + 9xy^2)) \neq \frac{\partial}{\partial x}(3x^2y - 5y^3)$$

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$$\frac{\partial}{\partial y}(-x^3 - 9xy^2) \neq \frac{\partial}{\partial x}(3x^2y - 5y^3)$$

$$-18xy \neq 6xy$$

and that answer choice D is not an exact equation.

$$\frac{\partial}{\partial y}(2x^2 + 9xy^2) \neq \frac{\partial}{\partial x}(3x^2y - 3y^3)$$

$$18xy \neq 6xy$$

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Topic: Exact equations

Question: Which values of a and b make the equation an exact differential equation?

$$(ay \sin x - \cos y)\, dx + (b \cos x + x \sin y)\, dy = 0$$

Answer choices:

A $(a, b) = (5,5)$

B $(a, b) = (-3, -5)$

C $(a, b) = (-5, -3)$

D $(a, b) = (5, -5)$

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Solution: D

If the equation is exact, then its partial derivatives are equal.

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\frac{\partial}{\partial y}(ay \sin x - \cos y) = \frac{\partial}{\partial x}(b \cos x + x \sin y)$$

$$a \sin x + \sin y = -b \sin x + \sin y$$

These partial derivatives are only equal when $a = -b$, which is only true for answer choice D.

Alternatively, we could also substitute each answer choice into the equation, and then apply the partial derivatives test. If we use the values from answer choice D, we'll plug $a = 5$ and $b = -5$ into the equation.

$$(5y \sin x - \cos y)\, dx + (-5 \cos x + x \sin y)\, dy = 0$$

Then we can use our partial derivatives test determine whether or not the equation is exact.

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\frac{\partial}{\partial y}(5y \sin x - \cos y) = \frac{\partial}{\partial x}(-5 \cos x + x \sin y)$$

$$5 \sin x + \sin y = 5 \sin x + \sin y$$

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Since these partial derivatives are equal, the differential equation must be exact when $a = 5$ and $b = -5$.

We can use the same process to show that answer choice A is not an exact equation,

$$(5y \sin x - \cos y)\, dx + (5 \cos x + x \sin y)\, dy = 0$$

$$\frac{\partial}{\partial y}(5y \sin x - \cos y) \neq \frac{\partial}{\partial x}(5 \cos x + x \sin y)$$

$$5 \sin x + \sin y \neq -5 \sin x + \sin y$$

that answer choice B is not an exact equation,

$$(-3y \sin x - \cos y)\, dx + (-5 \cos x + x \sin y)\, dy = 0$$

$$\frac{\partial}{\partial y}(-3y \sin x - \cos y) \neq \frac{\partial}{\partial x}(-5 \cos x + x \sin y)$$

$$-3 \sin x + \sin y \neq 5 \sin x + \sin y$$

and that answer choice C is not an exact equation.

$$(-5y \sin x - \cos y)\, dx + (-3 \cos x + x \sin y)\, dy = 0$$

$$\frac{\partial}{\partial y}(-5y \sin x - \cos y) \neq \frac{\partial}{\partial x}(-3 \cos x + x \sin y)$$

$$-5 \sin x + \sin y \neq 3 \sin x + \sin y$$

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Topic: Exact equations

Question: What is the solution to the exact differential equation?

$$(2xy + 2y)\,dx + (x^2 + 2x)\,dy = 0$$

Answer choices:

A $c = x^2y - 2xy - y$

B $c = x^2y + 2xy$

C $c = 2x^2y + 2xy + y$

D $c = x^2y + 4xy$

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Solution: B

We'll check to see that the equation is exact.

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\frac{\partial}{\partial y}(2xy + 2y) = \frac{\partial}{\partial x}(x^2 + 2x)$$

$$2x + 2 = 2x + 2$$

The functions $M(x, y)$ and $N(x, y)$ are equally easy to integrate, so we'll just use $M(x, y)$, and then Ψ can be given by

$$\Psi = \int M(x, y)\ dx + h(y)$$

$$\Psi = \int 2xy + 2y\ dx + h(y)$$

$$\Psi = x^2y + 2xy + h(y)$$

We'll differentiate both sides with respect to y,

$$\Psi_y = x^2 + 2x + h'(y)$$

and then substitute $\Psi_y = N(x, y)$ to solve for $h'(y)$.

$$x^2 + 2x = x^2 + 2x + h'(y)$$

$$0 = h'(y)$$

To find $h(y)$, we'll integrate both sides of this equation with respect to y.

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$$\int 0 \, dy = \int h'(y) \, dy$$

$$k_1 = h(y) + k_2$$

$$h(y) = k_2 - k_1$$

$$h(y) = k$$

Plugging this value for $h(y)$ into the equation for Ψ gives

$$\Psi = x^2y + 2xy + h(y)$$

$$\Psi = x^2y + 2xy + k$$

Setting $\Psi = c$ to find the solution to the exact differential equation, we get

$$c = x^2y + 2xy + k$$

$$c - k = x^2y + 2xy$$

$$c = x^2y + 2xy$$

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